

Project Summary

This project was developed as part of the AI Automations & Agent Foundations module. The objective was to design and implement an automated workflow that assists Volta Coffee Roasters in processing wholesale customer orders more efficiently while keeping a human decision-maker involved whenever necessary.

Wholesale orders are often submitted as free-text messages containing product requests, quantities, delivery details, and special instructions. Reading each message manually and transferring the information into a structured system is repetitive and can delay order processing. To address this challenge, an end-to-end workflow was created using n8n to automate the first stage of the order handling process.

The workflow begins when a customer submits an online order form. Google Gemini analyzes the order and extracts structured information such as products, quantities, delivery details, and customer requests. A JavaScript node validates and standardizes the AI output before the information is stored in Airtable. Based on the extracted information, the workflow classifies the order as either Standard, Needs Review, or Needs Clarification. A Switch node then routes the order to the appropriate Gmail node, which sends the corresponding customer response.

During development, particular attention was given to improving the reliability of the AI output. The Gemini prompt was refined to return a consistent JSON structure, while additional validation logic was added to handle missing values and standardize classification labels before the data reached Airtable. These improvements reduced workflow errors and increased consistency across different order scenarios.

The workflow was validated using three representative business scenarios: a complete order, an order requiring manual review because of a substitution request, and an incomplete order requiring clarification. In all cases, the workflow successfully extracted the relevant information, stored the order in Airtable, selected the correct workflow branch, and generated the expected customer email.

This project strengthened my understanding of workflow automation, prompt engineering, AI-assisted information extraction, JSON processing, API integration, and conditional workflow logic. It also highlighted the importance of combining AI with validation rules and human oversight to build reliable business automations.

Although the current workflow meets the project objectives, future improvements could include inventory checking, invoice generation, CRM integration, multilingual support, and manager approval workflows. Overall, the project demonstrates how AI and workflow automation can reduce repetitive administrative work while maintaining accuracy and human control over business decisions.